A CYANOTIC, DYSPNEIC ELDERLY MAN: RESPIRATORY FAILURE

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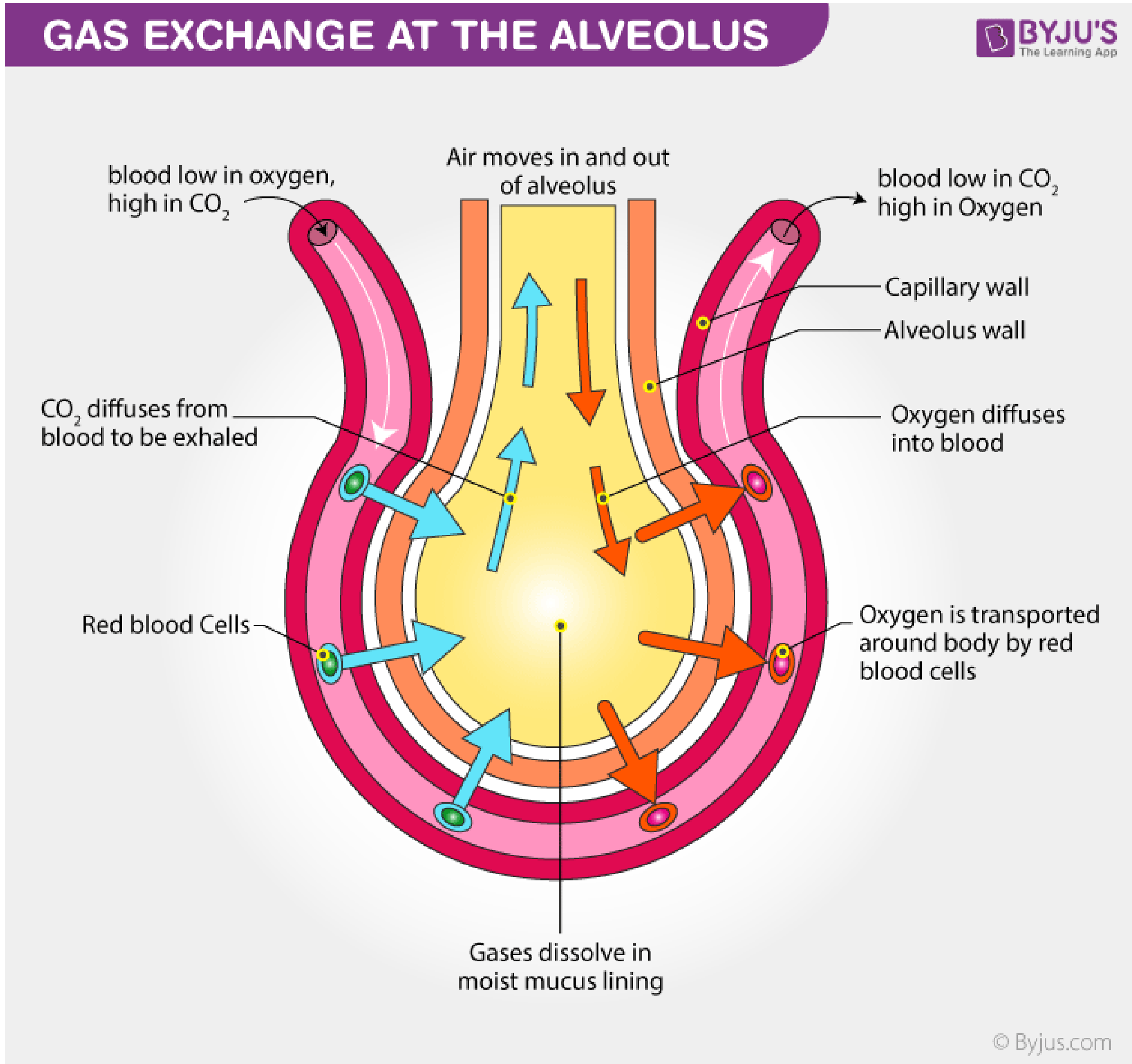
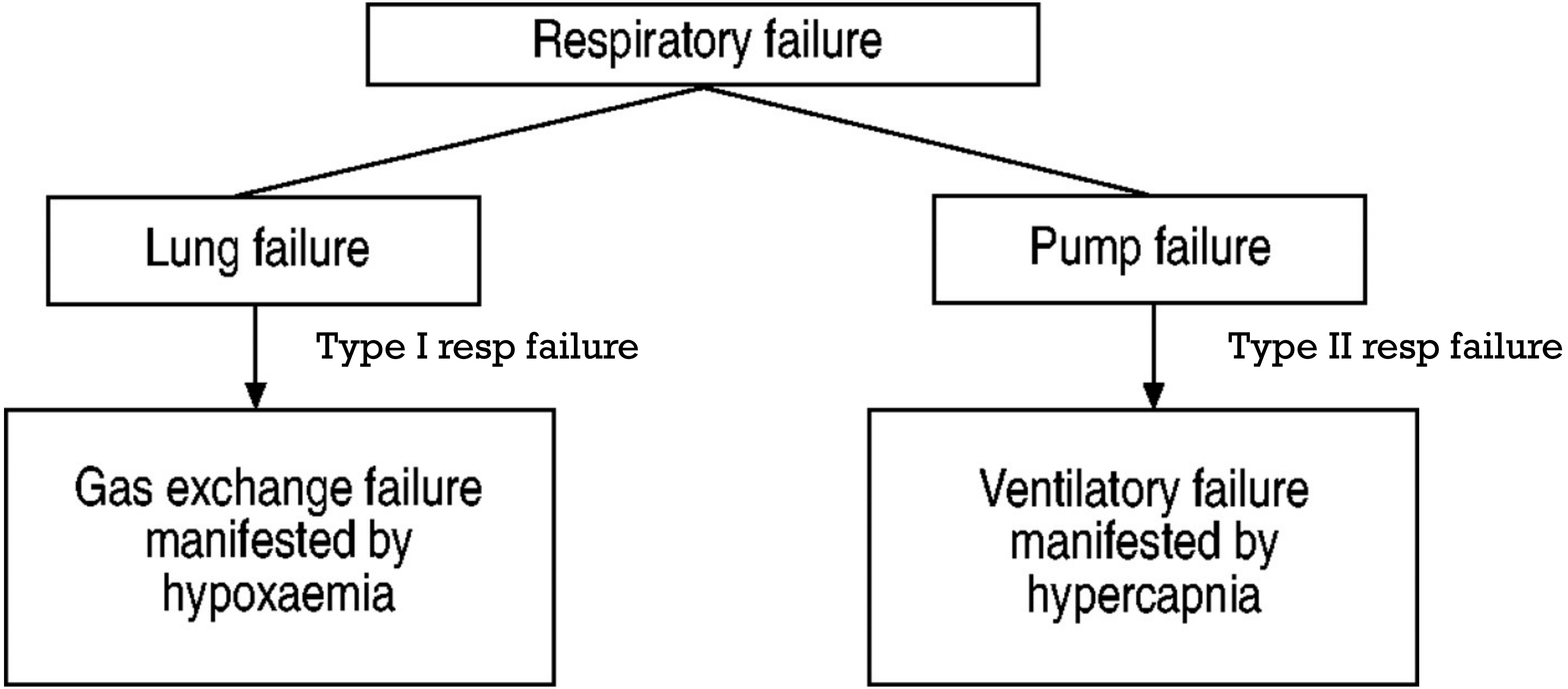
GENERAL CONCEPTS IN GAS EXCHANGE

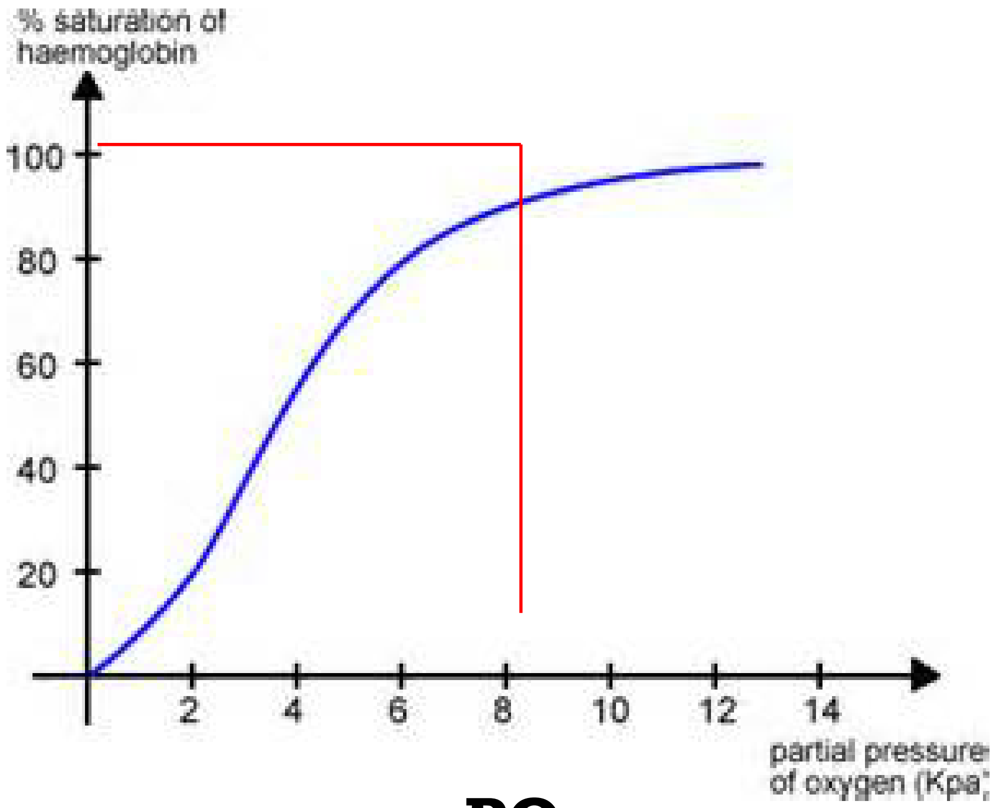


RESPIRATORY FAILURE



OXYGEN DESATURATION CURVE

SaO_2



PO_2

Hypoxemic respiratory failure: arterial ***$PO_2 \leq 8$ kPa***

HYPERCAPNIC RESPIRATORY FAILURE

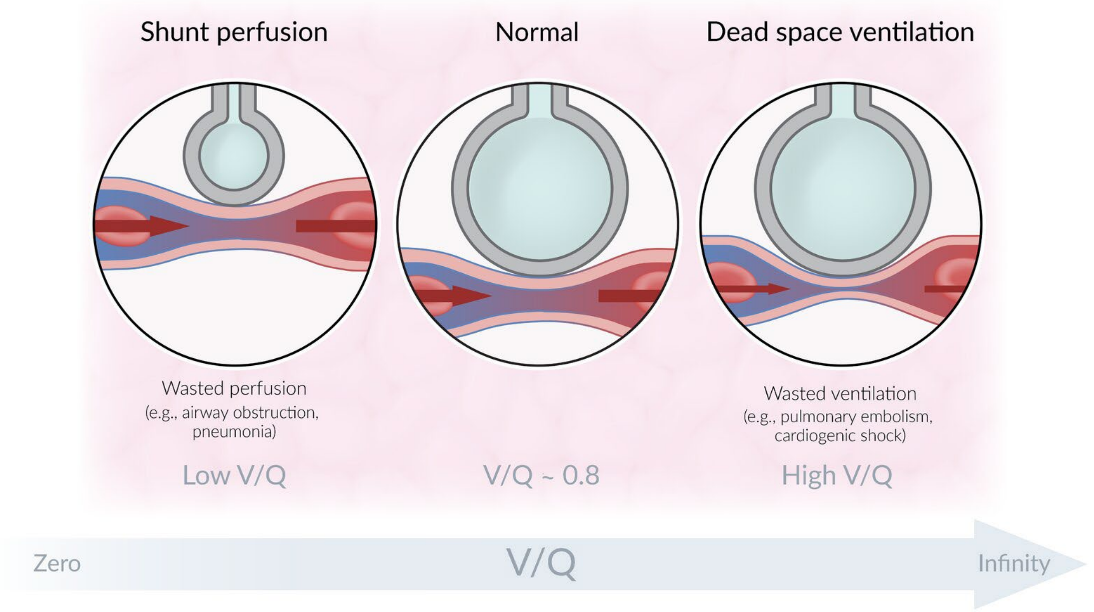
- Retention of CO_2 with acidosis (pH <7.30)
- Occur when the ventilation is unable to completely remove the produced metabolic wastes (CO_2)
- Arterial blood gases
  - Blood taken from artery
  - Send to the lab in ice as soon as possible
  - No excessive heparin

TYPE I RESP FAILURE

- Hypoxemia is defined as an abnormally low level of oxygen in the blood (ie, low partial oxygen tension)
- Hypoxia is defined as a condition where the oxygen supply is inadequate either to the body as a whole (general hypoxia) or to a specific region (tissue hypoxia)

TYPE I RESP FAILURE - MECHANISMS

- **V/Q mismatch**
  - Ventilation-perfusion (V/Q) mismatch refers to an imbalance of blood flow and ventilation
  - It causes the composition of alveolar gas to vary among lung regions:
    - Lung regions with low ventilation compared to perfusion will have a low alveolar oxygen content and high CO_2 content
    - Lung regions with high ventilation compared to perfusion will have a low CO_2 content and high oxygen content



TYPE I RESP FAILURE - MECHANISMS

- **V/Q mismatch**
  - In the diseased lung, V/Q mismatch increases because heterogeneity of both ventilation and perfusion worsen
  - The net effect is hypoxemia
  - Hypoxemia due to V/Q mismatch can be corrected with low to moderate flow supplemental oxygen

TYPE I RESP FAILURE - MECHANISMS

- **V/Q mismatch**
  - Obstructive lung diseases
  - Pulmonary vascular diseases
  - Interstitial lung diseases
  - Low flow/impaired cardiac output states (reduce perfusion globally including reducing perfusion to the pulmonary capillary bed
  - Physiologic dead space
    - Areas that are ventilated and not perfused; high V/Q
    - Large saddle pulmonary embolism, severe obstructive lung disease

TYPE I RESP FAILURE - MECHANISMS

- **Right-to-left shunt**
  - A right-to-left shunt exists when blood passes from the right to the left side of the heart without being oxygenated
  - Anatomic shunts exist when the alveoli are bypassed
    - Intracardiac shunts, pulmonary arteriovenous malformations (AVMs), and hepatopulmonary syndrome
  - Physiologic shunts exist when non-ventilated alveoli are perfused
    - Atelectasis and diseases with alveolar filling (eg, blood, pus, cells, water, or microbes)
  - Right-to-left shunts cause extreme V/Q mismatch, with a V/Q ratio of zero in some lung regions
  - The net effect is hypoxemia, which is difficult to correct with supplemental oxygen

TYPE I RESP FAILURE - MECHANISMS

- **Right-to-left shunt**
  - The degree of shunt can be quantified from the shunt equation
  - $Qs/Qt = (CcO_2 - CaO_2) \div (CcO_2 - CvO_2)$
  - Qs/Qt: Shunt fraction
  - CcO_2: End-capillary oxygen content
  - CaO_2 : Arterial oxygen content
  - CvO_2 is the mixed venous oxygen content

  - CaO_2 and CvO_2 are calculated from arterial and mixed venous blood gas measurements, respectively
  - CcO_2 is estimated from the PAO_2

TYPE I RESP FAILURE - MECHANISMS

- **Right-to-left shunt**
  - CaO_2 (mL O_2/dL) = (1.34 x hemoglobin concentration x SaO_2) + (0.0031 x PaO_2)
    - SaO_2 is the arterial oxyhemoglobin saturation
    - PaO_2 is the arterial oxygen tension
  - CvO_2 (mL O_2/dL) = (1.34 x hemoglobin concentration x SvO_2) + (0.0031 x PvO_2)
    - SvO_2 is the mixed venous oxyhemoglobin saturation
    - PvO_2 is the mixed venous oxygen tension
    - Normal CvO_2 is approximately 15 mL O_2/dL
    - Mixed venous blood is drawn from the right atrium

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TYPE I RESP FAILURE - MECHANISMS

- **Diffusion limitation**
  - Diffusion limitation exists when the movement of oxygen from the alveolus to the pulmonary capillary is impaired
  - It is usually a consequence of alveolar and/or interstitial inflammation and fibrosis, such as that due to interstitial lung disease
  - In such diseases, diffusion limitation usually coexists with V/Q mismatch, which makes the relative contribution of each to the patient's hypoxemia uncertain

TYPE I RESP FAILURE - MECHANISMS

- **Diffusion limitation**
  - Diffusion limitation is characterized by exercise-induced or -exacerbated hypoxemia
  - During rest, blood traverses the lung relatively slowly
    - Thus, there is usually sufficient time for oxygenation to occur even if diffusion limitation exists.

TYPE I RESP FAILURE - MECHANISMS

- **Diffusion limitation**
  - During exercise, cardiac output increases and blood traverses the lung more quickly
    - As a result, there is less time for oxygenation
    - In the healthy individuals, several compensatory mechanisms occur
      - Pulmonary capillaries dilate, which increases the surface area available for gas exchange by perfusing additional regions of lung
      - PAO_2 also increases, which promotes oxygen diffusion by increasing the oxygen gradient from the alveolus to the artery
      - The net effect is that full oxygenation is sustained
    - In patients with diffusion limitation (such as with pulmonary fibrosis), there is insufficient time for oxygenation to occur
      - In addition, most such patients have parenchymal destruction, which renders it impossible to recruit additional surface area for gas exchange
      - The net effect is measurable hypoxemia

TYPE I RESP FAILURE - MECHANISMS

- **Hypoventilation**
  - The lung alveolus is a space in which gas makes up 100% of the contents
  - Once the partial pressure of one gas rises, the other must decrease
  - Both arterial ($PaCO_2$) and alveolar ($PACO_2$) carbon dioxide tension increase during hypoventilation, which causes the alveolar oxygen tension (PAO_2) to decrease
  - As a result, diffusion of oxygen from the alveolus to the pulmonary capillary declines with a net effect of hypoxemia and hypercapnia
  - **The $paCO_2$ is elevated**

TYPE I RESP FAILURE - MECHANISMS

- **Causes of hypoventilation**
  - CNS depression, such as drug overdose, structural CNS lesions, or ischemic CNS lesions that impact the respiratory center
  - Obesity hypoventilation syndrome
  - Impaired neural conduction, such as amyotrophic lateral sclerosis, Guillain-Barré syndrome, high cervical spine injury, phrenic nerve paralysis, or aminoglycoside blockade
  - Muscular weakness, such as myasthenia gravis, idiopathic diaphragmatic paralysis, polymyositis, muscular dystrophy, or severe hypothyroidism
  - Poor chest wall elasticity, such as a flail chest or kyphoscoliosis

TYPE II RESP FAILURE - MECHANISMS

- The CO_2 level in arterial blood is directly proportional to the rate of CO_2 (VCO_2) production and inversely proportional to the rate of CO_2 elimination by the lung (alveolar ventilation)
- Alveolar ventilation (V_A) is, in turn, determined by minute ventilation (V_E) and the ratio of dead space (V_D) to tidal volume (V_T) ($V_A = V_E \times [1 - V_D/V_T]$)
- Increased dead space and reduced minute ventilation are common causes of hypercapnia

TYPE II RESP FAILURE - MECHANISMS

Etiologies and mechanism of hypercapnia

Respiratory pathway affecting carbon dioxide elimination	
Central nervous system ↓	**"Won't breathe"**
Peripheral nervous system ↓	**"Can't breathe"**
Respiratory muscles ↓	
Chest wall and pleura ↓	
Upper airway ↓	
Lungs	**Abnormal gas exchange: "Can't breathe enough"**

Schematic figure representing the respiratory pathway, along which a variety of diseases can affect carbon dioxide elimination and result in hypercapnia. Note that gas exchange abnormalities alone are relatively uncommon causes of hypercapnia, but gas exchange problems in the setting of reduced mechanical capability of the ventilatory pump are very common explanations for acute and chronic hypercapnia.

TYPE II RESP FAILURE

Mechanism and etiologies of hypercapnia

Mechanism	Etiologies		
Decreased minute ventilation (global hypoventilation; extra pulmonary causes)			
Decreased central respiratory drive	• Sedative overdose (eg, narcotic or benzodiazepine, some anesthetics, tricyclic antidepressants) • Encephalitis • Stroke • Central and obstructive sleep apnea • Obesity hypoventilation • Congenital central alveolar hypoventilation • Brainstem disease • Metabolic alkalosis • Hypothyroidism* • Hypothermia • Starvation		
Decreased respiratory neuromuscular or thoracic cage function	**Primary spinal cord/lower motor neuron/muscle disorders** • Cervical spine injury or disease (eg, trauma syringomyelia)[¶] • Amyotrophic lateral sclerosis • Poliomyelitis • Guillain-Barré syndrome • Phrenic nerve injury • Critical illness polymyoneuropathy • Myasthenia gravis • Muscular dystrophy • Polymyositis • Tetanus • Transverse myelitis (eg, multiple sclerosis) • Tick paralysis • Acute intermittent porphyria • Eaton Lambert syndrome • Neuralgic amyotrophy • Periodic paralysis • Glycogen storage and mitochondrial diseases • Respiratory muscle fatigue	**Thoracic cage disorders** • Kyphoscoliosis • Thoracoplasty • Flail Chest • Ankylosing spondylitis • Pectus excavatum • Fibrothorax	**Metabolic disorders**[Δ] • Hypophosphatemia • Hypomagnesemia • Hypothyroidism • Hyperthyroidism **Toxins, poisoning, drugs** • Tetanus • Dinoflagellate poisoning • Shellfish poisoning (red tide) • Ciguatera poisoning • Botulism • Organophosphates • Succinylcholine and neuromuscular blockade • Procainamide

TYPE II RESP FAILURE

Mechanism and etiologies of hypercapnia	
Increased dead space (gas exchange abnormalities; pulmonary parenchymal causes or airway disorders)	
Anatomic	Short shallow breathing
Physiologic	▪ Pulmonary embolism (usually severe) ▪ Pulmonary vascular disease (usually severe) ▪ Dynamic hyperinflation (eg, upper and lower airway disorders including chronic obstructive pulmonary disease, severe asthma) ▪ Endstage interstitial lung disease
Increased carbon dioxide production	
	▪ Fever ▪ Thyrotoxicosis ▪ Increased catabolism (sepsis, steroids) ▪ Overfeeding ▪ Metabolic acidosis ▪ Exercise
Multifactorial	
	Upper airway disorders◊ ▪ Severe laryngeal or tracheal disorders (stenosis/tumors/angioedema/tracheomalacia) ▪ Vocal cord paralysis ▪ Epiglottitis ▪ Foreign body aspiration ▪ Retropharyngeal disorders ▪ Obstructive goiter

EFFECTS OF HYPOXEMIA

- Cellular hypoxia is a state in which there is insufficient oxygen to meet the metabolic demands of a given tissue
- Cellular tolerance of hypoxia is variable
  - Skeletal muscle cells can recover fully after 30 minutes of hypoxia, but irreversible damage occurs in brain cells after only four to six minutes of similar hypoxic stress
- Cellular mechanisms that contribute to hypoxic cell injury
  - Depletion of ATP
  - Development of intracellular acidosis
  - Increased concentrations of metabolic by-products, generation of oxygen free radicals, and destruction of membrane phospholipids
  - Increase in intracellular calcium concentration, contributing to cellular injury via a variety of mechanisms, including direct damage to the cytoskeleton and induction of genes that contribute to apoptosis
  - Induces an inflammatory reaction characterized by neutrophilic infiltration, thus augmenting cellular damage via release of cytokine mediators, oxygen free radicals, and by intensifying ischemia due to disruption of the microcirculation

SYMPTOMS AND SIGNS OF RESPIRATORY FAILURE

- Respiratory distress
  - Tachypnea , respiratory rate >20/min (normal 12-14)
  - Tachycardia
  - Cyanosis
  - Use of accessory muscles (neck , intercostal in-sucking), gasping
- Drowsiness (hypoxia, acute CO_2 retention)
- Bradycardia, cardiac arrest

CLINICAL FEATURES OF HYPOXEMIA

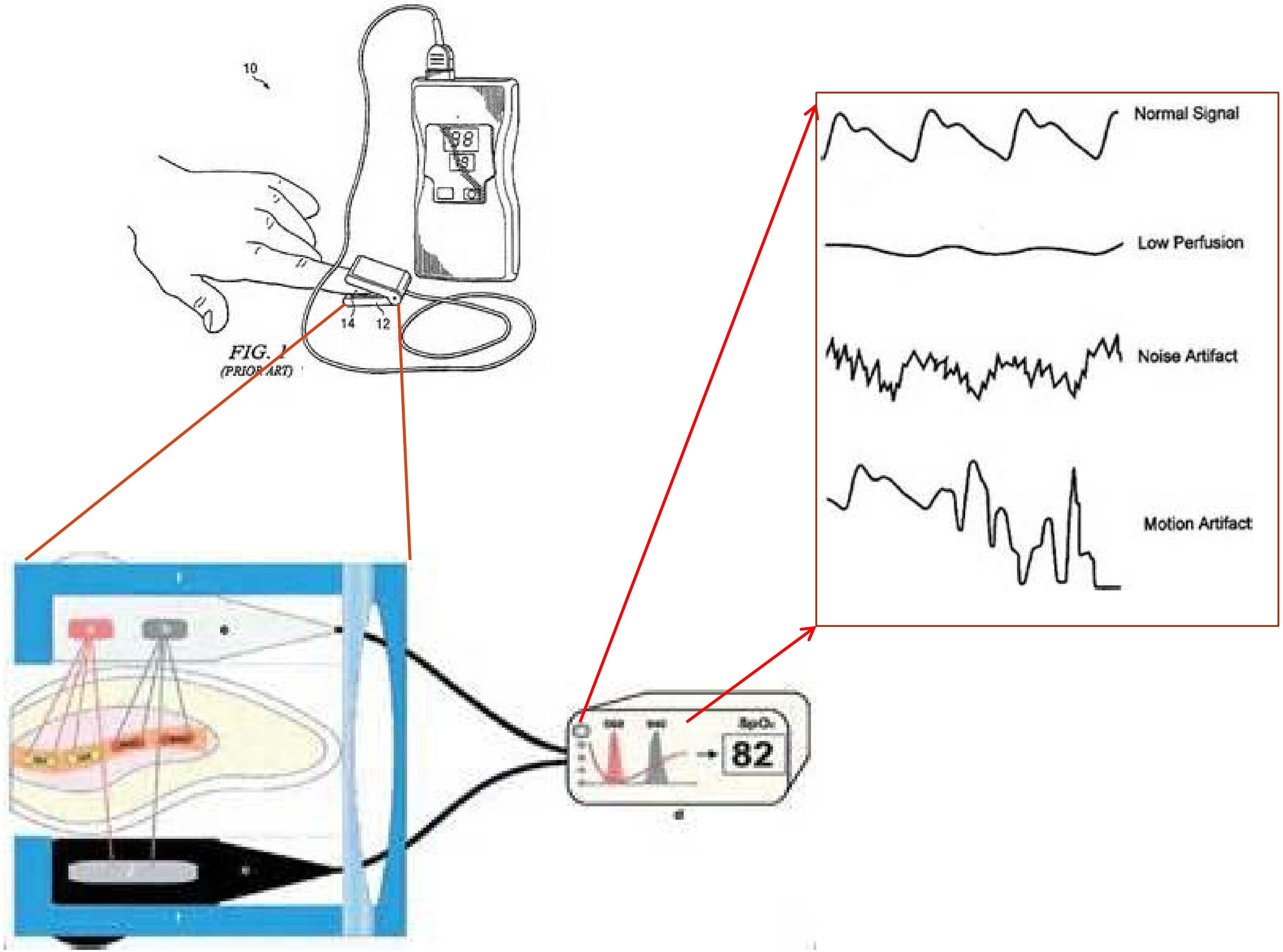
- Minor hypoxemia may cause few symptoms or signs
- More substantial desaturation can be accompanied by increased respiratory drive (therefore tachypnea and breathlessness), increased sympathetic tone, anxiety, and restlessness and sweating
- Hypoxemia produces arterial vasodilation and venous constriction (which is in part related with increased sympathetic tone).

MEASURES OF OXYGENATION

- **Arterial oxygen saturation (SpO2 and SaO2)**
- **Arterial oxygen tension (PaO2)**
- **A-a oxygen gradient**
  - The alveolar to arterial (A-a) oxygen gradient is a common measure of oxygenation ("A" denotes alveolar and "a" denotes arterial oxygenation)
  - It is the difference between the amount of the oxygen in the alveoli (ie, the alveolar oxygen tension [PAO_2]) and the amount of oxygen dissolved in the plasma (PaO_2)
  - A-a oxygen gradient = $PAO_2 - PaO_2$
    - $PAO_2 = (FiO_2 \times [Patm - PH_2O]) - (PaCO_2 \div R)$
    - FiO_2: Fraction of inspired oxygen (0.21 at room air)
    - Patm: Atmospheric pressure (760 mmHg at sea level)
    - PH_2O: Partial pressure of water (47 mmHg at 37°C)
    - $PaCO_2$: Arterial carbon dioxide tension
    - R: Respiratory quotient (0.8 at steady state)
  - A-a gradient = 2.5 + 0.21 x age in years

PULSE OXIMETRY

- A sensor is placed at perfused thin part of the body e.g. fingertip, earlobe
- Detect pulse rate (***important in determining signal quality***)
- Emit 2 wavelength (660nm, 940nm) red and infrared lights, determine the absorbances due to the pulsing arterial blood alone, excluding venous blood, skin, bone, muscle, and fat



10
14
12
FIG.
(PRIOR ART)
SpO2
82
Normal Signal
Low Perfusion
Noise Artifact
Motion Artifact

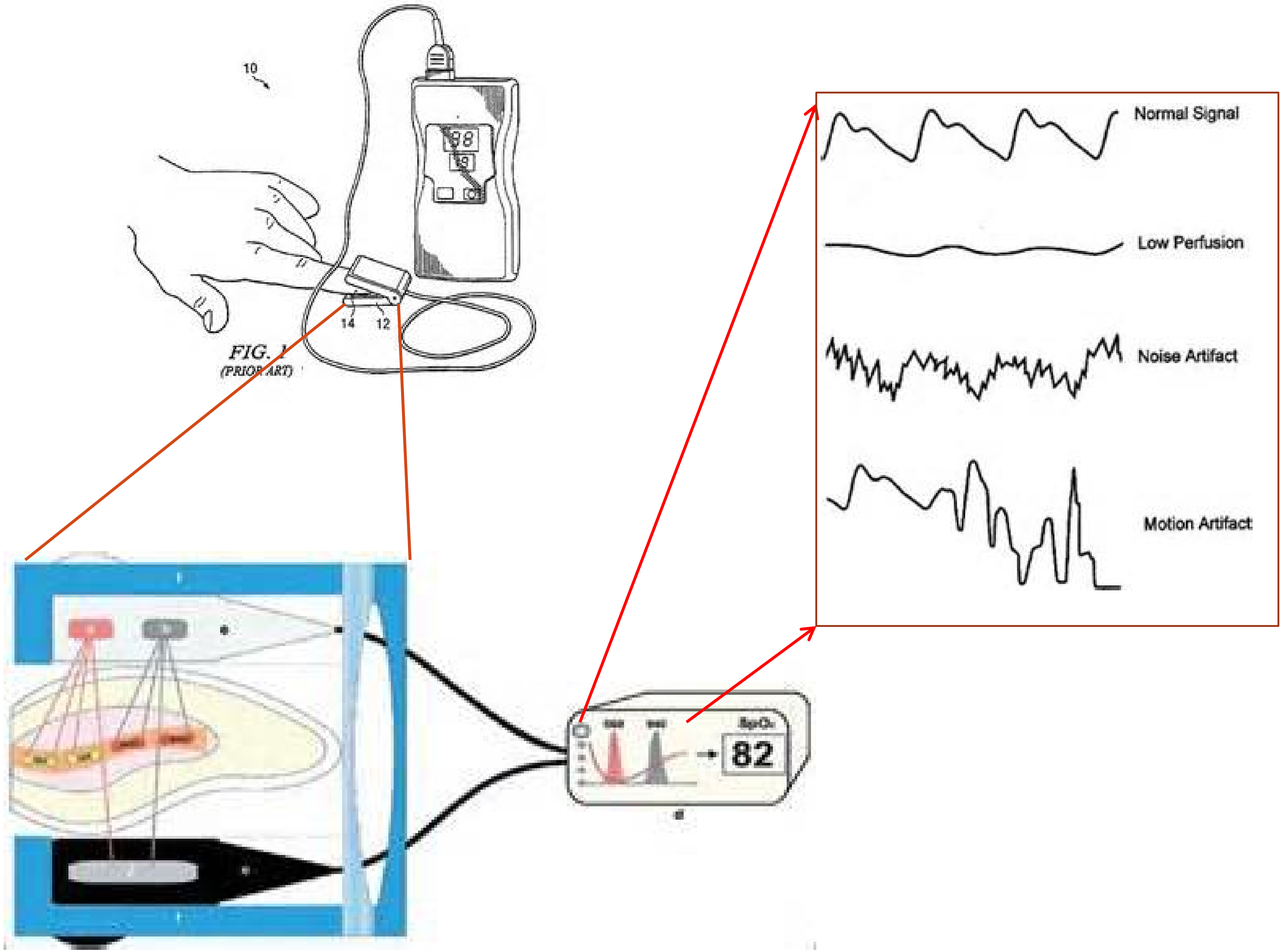

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FIG. 1
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Normal Signal
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Motion Artifact

A-A OXYGEN GRADIENT

Cause for hypoxia	A-a gradient	Corrected with high FIO2?	Causes
Low FIO2	Normal	Yes	High altitude, hypoxic gas mixture
Hypoventilation	Normal	Yes	Residual anesthetic, muscle relaxants
Diffusion	Elevated	Yes	Interstitial lung disease
V/Q mismatch	Elevated	Yes	Mucus plug, pulm embolism, COPD
Shunt	Elevated	No	Atelectasis, ARDS

MEASURES OF OXYGENATION

- **PaO2/FiO2 ratio**
  - The arterial oxygen tension/fraction of inspired oxygen (PaO_2/FiO_2) ratio
  - A normal PaO_2/FiO_2 ratio is 300 to 500 mmHg
  - <300 mmHg indicating abnormal gas exchange
  - <200 mmHg indicates severe hypoxemia

CLINICAL FEATURES OF HYPERCAPNIA

- Mild to moderate/develops slowly: Anxious, mild dyspnea, daytime sluggishness, headaches, or hypersomnolence
- Higher levels of CO_2 or rapidly developing - Frank alterations in sensorium including delirium, paranoia, depression, and confusion, which progress to somnolence and then coma (CO_2 narcosis) as levels continue to rise
- Severe - Asterixis, myoclonus, and seizures as well as papilledema, and dilated superficial veins

CLINICAL ASSESSMENT

- Mental status
- Heart rate
- Respiratory rate
- Respiratory pattern
  - Rapid shallow respiratory efforts, sternal retraction, use of accessory muscles
  - Thoracoabdominal paradoxical movement (the diaphragm moves cranially and the abdomen moves inward with inspiration) is a clinical indicator that the diaphragm is working against a fatiguing load.
  - Respiratory alterans, in which respiratory movements are abdominal for a few breaths, followed by a series of breaths in which respiration takes place predominantly by rib cage movements, is also considered as a sign of increased load
- Inspection of the skin, lips, tongue, and nail beds, to assess for cyanosis
- Pulmonary and cardiac auscultation
- Blood pressure, ECG
- Search for signs of cor pulmonale (peripheral edema, dilated jugular veins, hepato-jugular sign)

MANAGEMENT

- Assess airway, breathing, circulation
- Arterial blood gas analysis
- +/- Determining the alveolar-arterial gradient
- CXR
- +/- CT thorax/CT brain
- Administration of oxygen
- **+/- Ventilatory support**
- Treat the underlying cause

ABG INTERPRETATION

1. pH
   - Acidemic : pH <7.35
   - Normal : 7.35-7.45
   - Alkalemic : >7.45
2. pCO2 (N: 4-6 kPa)
   - If elevated with acidemia→ respiratory acidosis
3. HCO3- (N: 21-27 mEq/L)
   - Metabolic status
   - Renal compensation, chronicity

THERAPEUTIC TARGETS

- Reversion of underlying pathology and of contributing or precipitating factors
- Reversion of underlying pathophysiologic mechanisms (increased elastic or resistive load, atelectasis, etc)
- Oxygen therapy: arterial hypoxemia is the most life-threatening abnormality in RF, and increasing SaO_2 to 85%–90% should have the highest priority when managing ARF
- Reduce oxygen requirements: fever, agitation, overfeeding, vigorous respiratory activity, and sepsis can markedly decrease oxygen consumption
- Optimization of oxygen transport through interventions in cardiac output or red cell transfusion
- Avoidance of iatrogenic complications (overinflation, ventilator-induced lung injury, oxygen toxicity)

*RF indicates respiratory failure; ARF, acute respiratory failure.

†Modified from Greene and Peters,[6] Wood et al,[43] Calverley,[70] and Hall et al.[65]

OXYGEN THERAPY

- Room air contains about 21% of oxygen (FiO2 21%)
- Various modes of delivery
  - Nasal cannula
  - Venturi mask
  - Non-rebreathing mask

NASAL CANNULA

- Prescribe the oxygen flow rate: liter/minute
- Fraction of inspired oxygen (FiO2) ***varies*** with respiratory rate and effort (variable amount of air entrainment)

1LPM = 24% FIO2

2LPM = 28% FIO2

3LPM = 32% FIO2

4LPM = 36% FIO2

5LPM = 40% FIO2

6LPM = 44% FIO2

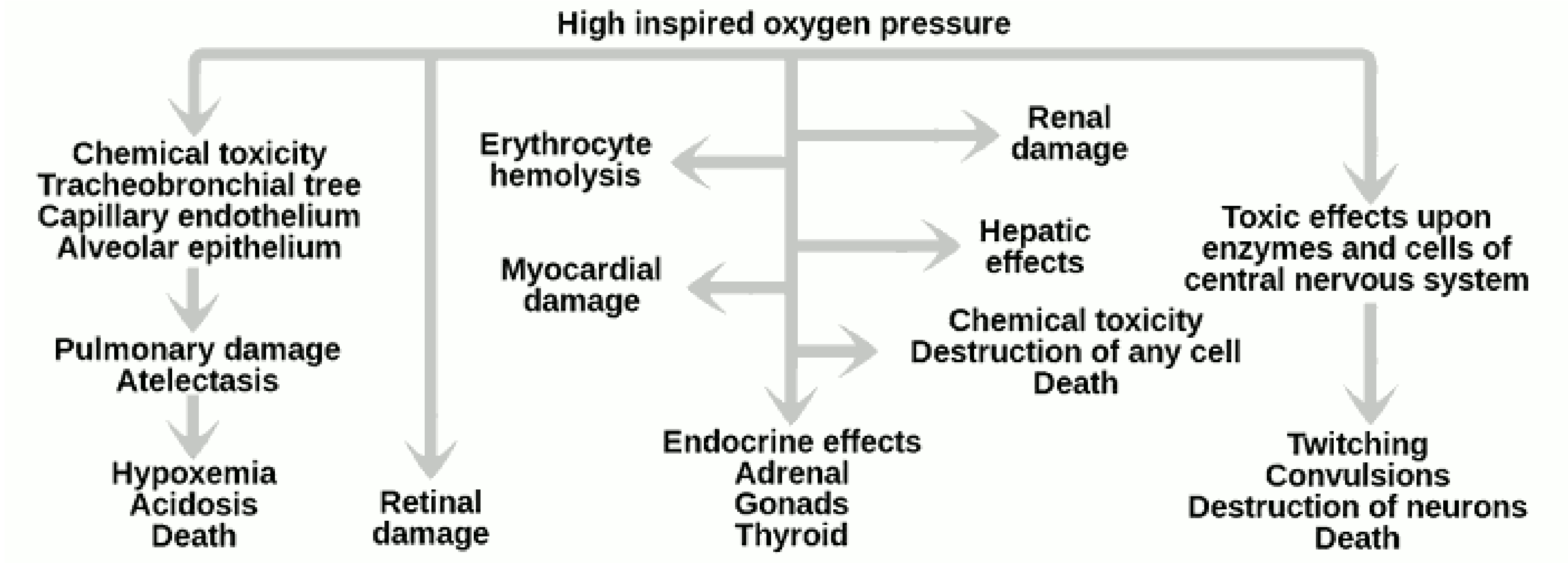
NASAL CANNULA

- Pros
  - Convenient, less disfigurement
  - Easy to feed, allow expectoration of sputum
- Cons
  - FiO2 not exact
  - ***High flow O2 (i.e. > 6 l/min)not recommended***, due to the discomfort, drying of nasal passage and epistaxis (humidification needed)
  - Dislodgement

SIDE EFFECTS OF OXYGEN THERAPY

- Carbon dioxide retention, especially in COPD (excessive oxygen knocks off the hypoxic respiratoy drive)
- Lung toxicity, exacerbate lung injury
- Oxidative stress and cell injury
- Resorption atelectasis

RISKS OF OXYGEN THERAPY



MEASURES TO IMPROVE ALVEOLAR VENTILATION

- Bronchodilation
- Control of infection and secretions
- Stabilization of the chest wall
- Careful control of O_2 therapy
- Avoidance of drugs that depress breathing or reversal of their effects
- NIPPV

*NIPPV indicates noninvasive positive pressure ventilation.

†Modified from Nunn[40] and Hall et al.[65]

INDICATIONS FOR NIPPV IN PATIENTS WITH TYPE 2 RF IN THE ACUTE SETTING

- Increasing dyspnea of at least medium severity
- More than 24 breaths per minute, use of accessory muscles, respiratory paradox
- $PaCO_2 < 45$ mm Hg, pH < 7.35
- $PaO_2/FiO_2 < 200$

*NIPPV indicates noninvasive positive pressure ventilation; RF, respiratory failure.

NIV IN THE ACUTE SETTING

- Strong evidence (multiple controlled trials)
  - **COPD**
  - **Acute cardiogenic pulmonary oedema**
  - **Immunocompromised patients with diffuse lung infiltrates**
- Less strong evidence (single controlled trial or multiple case series)
  - Pneumonia
  - Asthma
  - Postoperative respiratory failure
  - Avoidance of extubation failure
  - "Do not intubate" patients
- Weak evidence (few case series or case reports)
  - Upper airway obstruction
  - Acute respiratory distress syndrome
  - Trauma

NIV IN THE CHRONIC SETTING

- NPPV is also indicated as a long-term ventilatory support for patients suffering from chronic respiratory failure secondary to restrictive lung disease due to neuromuscular or chest wall disease, sleep-related breathing disorders and COPD

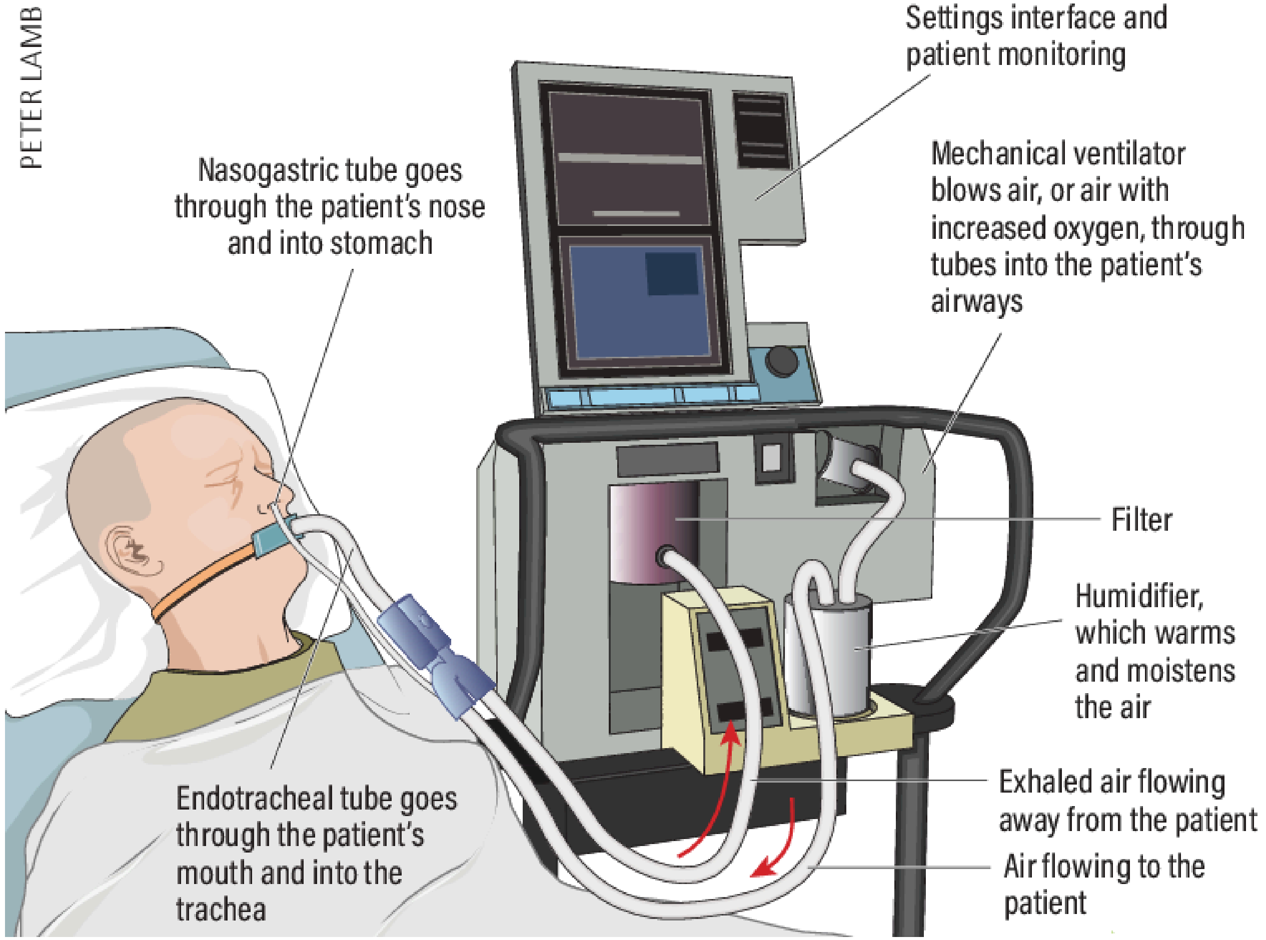
CONTRAINDICATIONS FOR NIV

- Cardiac/respiratory arrest
- Organ failure: (*a*) severe hemodynamic or electric instability, (*b*) severe encephalopathy (GCS score <10), and (*c*) acute gastrointestinal bleeding
- Acute coronary syndrome (unstable angina or myocardial infarction)
- Uncooperative patient
- Impaired consciousness. Nevertheless, patients with obtundation because of hypercapnia, often respond to NIPPV
- Severe bulbar paralysis (less severe disorder is only a relative contraindication)
- High risk for aspiration (vomiting, ileus)—need for airway protection
- Anatomic abnormalities of the nasopharynx, facial trauma or surgery, recent surgery of the upper airway
- Recent surgery in the upper gastrointestinal tract is a relative contraindication
- Fixed stenosis of the upper airway
- Inability to clear secretions

*NIPPV indicates noninvasive positive pressure ventilation; GCS, Glasgow Coma Scale.

VENTILATORY SUPPORT – NON-INVASIVE

VENTILATORY SUPPORT –INVASIVE



TREATMENT OF CHRONIC RF

- Treatment of underlying disease
- LTOT
- Rehabilitation
- Domiciliary mechanical ventilation (invasive or noninvasive)
- In selected patients with emphysema, lung-volume–reduction surgery
- Lung transplantation

*CRF indicates chronic respiratory failure; LTOT, long-term oxygen treatment.

INDICATIONS FOR LTOT IN COPD

- Resting $PaO_2 < 55$ mm Hg (or $SaO_2 < 89\%$)
- Resting $PaO_2 = 55–59$ mm Hg (or SaO_2 89%), in the presence of (*a*) cor pulmonale (*b*) polycythemia (hematocrit $> 56\%$)

*LTOT indicates long-term oxygen treatment; COPD, chronic obstructive pulmonary disease.